

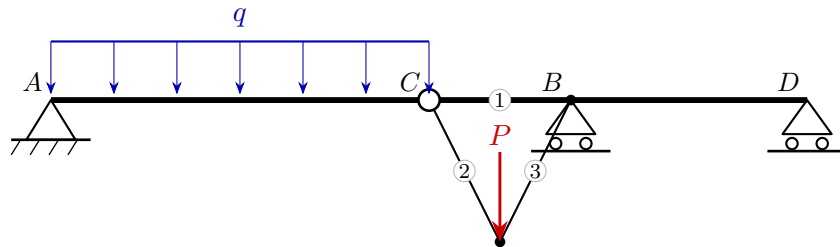
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam Group A

Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

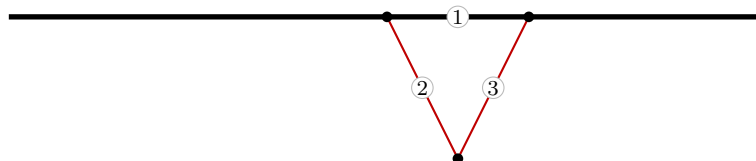
$$A: R_x = 0, R_y = 18 \quad D: R_x = 0, R_y = 8 \quad B: R_x = 0, R_y = 6$$

$$C: H_x = 0, H_y = -6$$

3. Truss member forces

Member	Force S [kN]	Type
1	-3	compression
2	6.71	tension
3	6.71	tension

red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams

